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Evaluation of notional nozzle approaches for CFD simulations of free-shear under-expanded hydrogen jets

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ABSTRACT

Several approaches are usually applied for modelling the source of high pressure under-expanded jets, ranging from the computationally expensive resolution of the jet's shock structure to simple formulae (pseudo-source or notional nozzle approaches). However, the assumptions made in each approach introduce inaccuracies in the CFD calculations. The objective of this work was twofold; to compare and evaluate the performance of both selected notional nozzle approaches and turbulence models with experimental results of free-shear high momentum H_2 round jets. The experimental data covered horizontal H_2 releases issuing from small nozzles (0.25–1 mm diameter). Three two-equation turbulence models were chosen for the simulations, the popular standard k-ε, the Shear Stress Transport (SST) and the baseline (BSL) k-ω model together with five notional nozzle approaches. The numerical results were presented in a systematic way in order to make general conclusions on the performance of both the approaches and models.

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1. Introduction

The accidental release from high pressure H_2 systems may lead to a jet of several meters size. Understanding and predicting the jet behaviour is an essential step in any safety analysis of high pressure systems. The jet will be under-expanded close to the release and will rapidly accelerate and expand to atmospheric pressure through a series of shocks. Numerical simulations to rigorously resolve the shock region with a satisfactory grid resolution are demanding. Together with the need for a long enough computational domain to encompass the subsonic flow further downstream, the numerical effort can be intensive. For example, in [1] a computational mesh of 200.000 nodes was used to model only ¼ of the shock region of a H_2 release from a 20 MPa vessel. Also, in HYPER [2] a grid resolution between 1/32 to 1/64 of the actual leak diameter was necessary to ensure grid independence. To overcome the expensive computational demands, several approaches have been proposed in the past which replace the actual nozzle by a notional one (often referred to as fictitious or pseudo-diameter nozzle) that occupies a larger area but with the same flow rate as the real one and at ambient pressure and uniform velocity. The approaches have been implemented in several numerical investigations. In [3] Birch1984 [4] approach was used to model horizontal H_2 releases of several high pressure experiments. The authors reported good agreement with the experimental measurements. In [5] the majority of the participants used notional

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nozzle approaches to model a 35 MPa pressure H_2 release. In [6] a notional nozzle approach was used to model several H_2 and CH_4 releases from compressed gaseous systems at 20, 35 and 70 MPa. In [7] a notional nozzle approach was used to model the release of CH_4 at pressures ranging from 0.85 to 10 barg inside a mechanically ventilated room and reported good predictions. In [8] numerical results to evaluate the LFL extent of vertical free H_2 and CH_4 with Birch1984 [4] and Schefer 2007 [9] notional nozzle approaches were presented. A comparison between the results of the actual and a notional nozzle approach (similar to Birch1984) of a H_2 release at 430 bar was presented in [10] and it was found that the notional approach produced 25–30% longer extents of the flammable cloud. The difference was attributed to the use of real gas properties in the actual leak modelling in contrast to the ideal gas properties and the different input sound velocity of the jet of the notional approach. Finally, the validation of notional nozzle approaches has been identified [11,12] as an issue that needs investigation.

2. Notional nozzle approaches

Some of the most known notional nozzle approaches are Birch1984 [4], Birch1987 [5], Ewan [13] and Schefer [9]. The Birch1984 approach is based on conservation of mass only and the Birch1987 on conservation of both mass and momentum. Both of them consider that the temperature of the jet is equal to atmospheric. Ewan's approach is the same as the Birch1987 but with the assumption that the temperature of the jet is equal to the one derived from the isentropic expansion of the jet from stagnant conditions to the actual nozzle. Schefer approach is analogous to Birch1987, but proposes the use of the Abel-Noble equation of state for the calculation of real gas properties instead of the ideal gas law. Attempts to propose a more accurate modelling of the leak source take also into account the conservation of energy, assuming that the jet is either ideal gas [14] or with properties being calculated using fundamental equations of state [15]. Furthermore, Harstad [16] proposed the location of the notional nozzle being just after the Mach disk with an area equal to that of the disk while keeping the assumption that the pressure is atmospheric.

3. Experimental description

Experiments of horizontal high momentum H_2 releases were carried out at the HYKA test site of the Institute for Nuclear and Energy Technologies of the FZK/KIT to evaluate the H_2 distribution and flow field in free turbulent jets at different conditions. The release orifice was located 0.9 m from the ground to avoid surface effects. The jets were released from a bulk vessel at initial pressures ranging from 20 to 250 bars and from nozzle diameters ranging from 0.25 mm to 1 mm into still air. Measurements of radial and centreline distributions of H_2 concentration and flow velocity were taken at 0.75 m, 1.5 m and 2.25 m from the nozzle. A description of the experiments can also be found in [17] and [18]. Three experimental sets were selected for this study, namely HD35-37, HD22-24 and HD00-02. The conditions of the experiments are shown in Table 1. The last column shows the location of the above mentioned measurements for each experimental condition.

4. Simulations set up

Table 2 presents the release conditions based on the different notional nozzle approaches. An average stagnation pressure (53.27 bar) was used for the HD35-37 experimental set, 2 stagnation pressures, i.e. 113.3 bar and the average of 104.8 bar and 99.6 bar (102.2 bar) for the HD00-02 set and finally a stagnation pressure equal to 162 bar was used for the HD22-24 set as initial conditions. The velocity of Birch1984 and Ewan approaches is sonic, the one of Birch1987 and Schefer is higher as it is assumed that all excess pressure goes to increasing the jet's momentum whereas the one of Harstad is far less than sonic as the conditions after the Mach disk are considered. The table presents also the momentum flow rate, the enthalpy flux and the net rate of energy flow (sum of specific enthalpy and specific kinetic energy) through each notional nozzle. The same parameters are given for the real nozzle based on the experimental conditions (assuming real gas properties and isentropic expansion at the nozzle) for comparison.

The numerical calculations were performed with ANSYS-CFX version 12.1 [19]. The following set up was applied to all simulations:

- Simulations were performed for 10 s to ensure steady state.
- The High Resolution scheme was used for discretization of advection terms (a blending factor allows a switch to 1st Order Advection scheme based on local variable gradients).
- The 2nd Order Backward Euler discretization scheme was used for transient terms.
- Total energy option was selected (transport of enthalpy including kinetic energy effects).
- The ideal gas law was used as equation of state.
- Stagnant conditions were assumed as the initial state with atmospheric pressure and temperature equal to 14.5 °C.
- The velocity and temperature, as calculated from the notional nozzle approaches (Table 2), were specified at the inlet. All simulations had a common incoming turbulence intensity (5%) assigned to the inlet. The effect of turbulence intensity at the source was investigated by increasing

Table 1 – Experimental conditions.

Test No.	Nozzle diameter (mm)	Pressure (bar)	Flow rate (10^{-3} kg/s)	Distance from nozzle (m)
HD35-37	1	54.6	2.46	0.75
		52.1	2.35	1.5
		53.1	2.39	2.25
HD00-02	0.75	113.3	2.87	0.75
		104.8	2.66	1.5
		99.6	2.53	2.25
HD22-24	0.25	162.8	0.46	0.75
		160.4	0.45	1.5
		162.1	0.46	2.25

Table 2 – Conditions at the notional nozzle.

Approach	Temperature (K)	Density (kg/m^3)	Velocity (m/s)	Diameter (10^{-3} m)	Mass flow rate (10^{-3} kg/s)	Momentum flow rate (kg m^2/s)	Enthalpy flux (kJ/s)	Net rate of energy flow (10^3 kJ/s)
HD35-37 (1 mm nozzle diameter, 53.27 bar)								
Birch1984	287.6	0.0854	1292	5.262	2.4	3.1	10.2	2.01
Birch1987	287.6	0.0854	1984	4.247	2.4	4.76	10.2	4.73
Ewan	238.8	0.1029	1178	5.024	2.4	2.83	8.58	1.67
Schefer	287.6	0.0854	2001	4.228	2.4	4.81	10.2	4.82
Harstad	278.4	0.0882	510	8.241	2.4	1.23	9.92	0.32
Exp. data	237	2.74	1200	1.0	2.4	2.88	57.2	1.78
HD00 (0.75 mm nozzle diameter, 113.3 bar)								
Birch1984	287.6	0.0854	1292	5.754	2.87	3.71	12.2	2.41
Birch1987	287.6	0.0854	1999	4.626	2.87	5.74	12.2	5.75
Ewan	238.8	0.1029	1178	5.492	2.87	3.38	10.3	2.00
Schefer	287.6	0.0854	2036	4.584	2.87	5.85	12.2	5.96
Harstad	278.6	0.0882	503	9.079	2.87	1.44	11.9	0.37
Exp. data	235	5.59	1221	0.75	2.87	3.50	133	2.27
HD01-02 (0.75 mm nozzle diameter, 102.2 bar)								
Birch1984	287.6	0.0854	1292	5.471	2.59	3.35	11.1	2.18
Birch1987	287.6	0.0854	1998	4.399	2.59	5.19	11.1	5.19
Ewan	238.8	0.1028	1178	5.223	2.59	3.06	9.27	1.81
Schefer	287.6	0.0854	2031	4.363	2.59	5.27	11.1	5.36
Harstad	278.6	0.0882	504	8.626	2.59	1.31	10.7	0.34
Exp. data	235	5.08	1217	0.75	2.59	3.16	110	2.03
HD22-24 (0.25 mm nozzle diameter, 162 bar)								
Birch1984	287.6	0.0854	1292	2.303	0.46	0.59	1.96	0.39
Birch1987	287.6	0.0854	2004	1.849	0.46	0.92	1.96	0.93
Ewan	238.8	0.1028	1178	2.199	0.46	0.54	1.64	0.32
Schefer	287.6	0.0854	2055	1.827	0.46	0.95	1.96	0.97
Harstad	278.7	0.0881	500	3.645	0.46	0.23	1.90	0.06
Exp. data	234	7.76	1239	0.25	0.46	0.57	29.4	0.38

its value from 5% (the "default" value) to 10% for the experimental set HD22-24 using Schefer approach and all 3 turbulence models and was found insignificant. The importance of the turbulence intensity has also been investigated in the past [20–22]. In [20] for example, the authors concluded that for moderate grid resolution, the turbulence intensity has a weak effect on the predictions, provided that its value is less than 30%. Also in [21] RANS simulations showed small sensitivity to inlet turbulence intensity

- The dimensions of the computational domain were 15 m × 10 m × 10 m. Ground was defined as wall (no slip boundary condition), west and top planes as openings (relative total pressure for inflow and relative static pressure for outflow set to zero, zero gradient for turbulent quantities), east plane as outlet (averaged relative static pressure set to zero) and finally for north and south planes a symmetry assumption (zero normal velocity component and zero normal scalar variable gradients) was made.
- The mesh used was unstructured. Information about the mesh characteristics is given in paragraph "Grid sensitivity analysis".

5. Non dimensional features of turbulent free-shear round jets

The configuration and coordinate system of a turbulent round jet is shown in Fig. 1.

In the self-similar region ($x/d_j > 30$) of high-Reynolds-number turbulent jets ($\mathrm{Re} > 10^4$, where $\mathrm{Re} = U_j d_j/\upsilon$), the centreline velocity $U_o(x) \equiv \langle U(x,0,0)\rangle$ and half width $r_{1/2}(x)$ (such that $\langle U(x, r_{1/2}(x), 0)\rangle = \frac{1}{2}U_0(x)$) vary according to [23]:

$$\frac{U_0(x)}{U_j} = \frac{B}{\frac{(x - x_0)}{d_j}} \tag{1}$$

$$r_{1/2}(x) = S(x - x_0) \tag{2}$$

Where B and S are empirical constants and x_o the virtual origin.

Pope [23] considers that the constants B and S of these laws are independent of Re and that their values are 5.8 and 0.094 respectively. In the case of fully turbulent ($Re > 3 \cdot 10^4$ [24]) buoyant jets the character of the flow is determined by the ratio of inertial to buoyant forces, the Froude number, which assuming uniform velocity and density profiles at the source is:

$$Fr = \frac{\rho_j U_j^2}{g d_j |\rho_\infty - \rho_j|} \tag{3}$$

A high Fr number leak ($Fr > 1000$ according to [25]) is momentum dominated whereas a low Fr number leak ($Fr < 10$ according to [25]) is dominated by buoyancy forces. Three regions can be observed in turbulent buoyant jets: the one close to the nozzle (momentum-dominated regime), followed by an intermediate region and the buoyancy-dominated regime in the far field. Similarly to the turbulent non

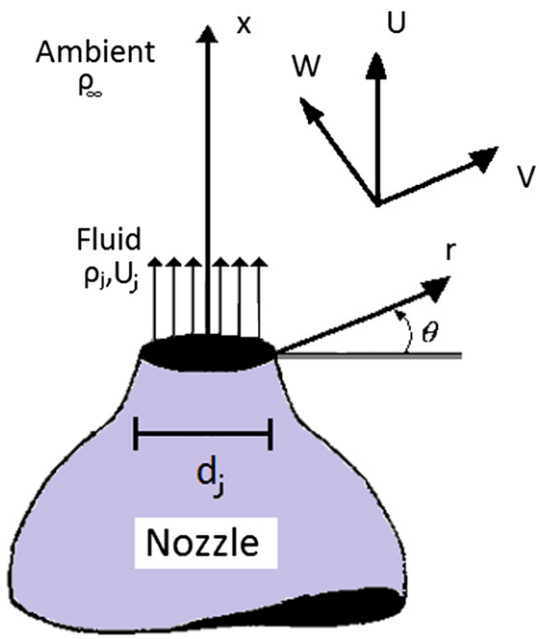

Fig. 1 – Schematic of a round-jet, showing the coordinate system (source).

buoyant round jet, Chen and Rodi [26] by reviewing experimental data of steady low Mach number vertical buoyant jets suggested that two distinct forms of self-similarity are possible, one in each of the momentum- and buoyancy-dominated regimes. In these similarity regions the variation of velocity and excess temperature (or concentration) along the centreline is different.

Chen and Rodi [26] proposed a scaled distance x_b to estimate the distance from the release at which buoyancy forces become important:

$$x_b = Fr^{-1/2}\left(\frac{\rho_j}{\rho_\infty}\right)^{-1/4}\left(\frac{x}{dj}\right) \tag{4}$$

They suggested that the flow is momentum-dominated when $x_b < 0.53$. However, Papanicolaou and List [27] showed that momentum dominance holds for longer distances from the leak ($x_b < 1$). Chen and Rodi [26] proposed the following decay laws for centreline velocity and volumetric concentration in a round buoyant jet (in the non-buoyant region):

$$\frac{U_c}{U_j} = 6.2\left(\frac{\rho_j}{\rho_\infty}\right)^{1/2}\left(\frac{x}{dj}\right)^{-1} \tag{5}$$

$$\frac{C_c}{C_j} = 5\left(\frac{\rho_j}{\rho_\infty}\right)^{-1/2}\left(\frac{x}{dj}\right)^{-1} \tag{6}$$

It is interesting to mention that the constant in eq. (5) has been reported by a recent review on turbulent jets [28] to vary from 5 to 6.7. Also, in [29] a similar formula to eq. (6) was proposed for mass fraction decay law for compressible H_2 flows but with a constant equal to 5.4 instead of 5. In [14] an analysis and experimental data to study the spreading and centreline property decay laws of under expanded supersonic jets (dried air with jet exit-to-ambient pressure ratios ranging from 1 to 20.3) was presented. The authors proposed the axial distance and velocity to be normalized by $(\rho_2/\rho_\infty)^{1/2}d^*$ and U_2 respectively (where ρ_2, d^* and U_2 are the density, "fictitious" diameter and velocity after the expansion region). In [14], it is stated that a closer grouping of velocity decay data of several sonic jets was achieved when they used the normalization strategy with $(\rho_2/\rho_\infty)^{1/2}d^*$ and U_2. For the same reason, the same type of normalization has been also used by other authors [15,17,18,21,30]. Because of the above reasons, we followed Yüceil and Ötügen' approach for the calculations of the normalized centreline flow velocity and H_2 mass fraction in the following paragraphs. In [14], the average value of the slope of the centreline velocity decay law was found equal to 6.25 whereas the total relative spread between the minimum and maximum slope of their experimental data was about 18%. They also noticed that the value of the slope is decreasing with increasing jet exit-to-ambient pressure ratios. However, they argued that no further decrease should be expected for higher than 20.3 ratios.

In Tables 3 and 4 the exit *Re* and *Fr* numbers of the experiments (assuming isentropic expansion from stagnant conditions to the nozzle, allowing for H_2 real gas behaviour) and notional nozzle approaches, together with the conservative distance of the momentum-dominated regime (i.e. for $x_b < 0.5$, from formula (4)) are shown. The table shows that the

Table 3 – Jet exit characteristics: Froude and Reynolds number at the nozzle/exit, distance x at which buoyancy becomes dominant.

		Exp.	Birch 1984	Birch 1987	Ewan	Schefer	Harstad
HD35-37 (1 mm)	*Re*	$1.16\cdot10^4$	$6.69\cdot10^4$	$8.29\cdot10^4$	$5.82\cdot10^4$	$8.33\cdot10^4$	$4.14\cdot10^4$
	Fr	$2.45\cdot10^8$	$2.48\cdot10^6$	$7.25\cdot10^6$	$2.64\cdot10^6$	$7.41\cdot10^6$	$2.56\cdot10^5$
	x (m)	9.71	2.14	2.95	2.21	2.97	1.09
HD00 (0.75 mm)	*Re*	$8.70\cdot10^3$	$7.32\cdot10^4$	$9.11\cdot10^4$	$6.37\cdot10^4$	$9.19\cdot10^4$	$4.49\cdot10^4$
	Fr	$2.35\cdot10^8$	$2.27\cdot10^6$	$6.76\cdot10^6$	$2.42\cdot10^6$	$7.07\cdot10^6$	$2.26\cdot10^5$
	x (m)	8.63	2.24	3.11	2.31	3.15	1.12
HD01-02 (0.75 mm)	*Re*	$8.70\cdot10^3$	$6.96\cdot10^4$	$8.65\cdot10^4$	$6.06\cdot10^4$	$8.73\cdot10^4$	$4.28\cdot10^4$
	Fr	$2.47\cdot10^8$	$2.39\cdot10^6$	$7.10\cdot10^6$	$2.54\cdot10^6$	$7.40\cdot10^6$	$2.38\cdot10^5$
	x (m)	8.46	2.18	3.03	2.25	3.07	1.10
HD22-24 (0.25 mm)	*Re*	$2.90\cdot10^3$	$2.93\cdot10^4$	$3.65\cdot10^4$	$2.55\cdot10^4$	$3.70\cdot10^4$	$1.79\cdot10^4$
	Fr	$6.57\cdot10^8$	$5.67\cdot10^6$	$1.70\cdot10^7$	$6.03\cdot10^6$	$1.81\cdot10^7$	$5.55\cdot10^5$
	x (m)	5.26	1.42	1.97	1.46	2.01	0.707

experimental measurements, taken at 0.75 m, 1.5 m and 2.25 m from the leak, fall well within the momentum-dominated regime. The extent of this region is not the same for the notional nozzle approaches. Obviously, Harstad approach creates a region much smaller than the rest, whereas Schefer followed by Birch1987 approach create the longer regions.

6. Results and discussion

The evaluation of the notional nozzle approaches, together with the 3 turbulence models, was mainly based on their ability to predict the centreline H_2 mass fraction and flow velocity at the three measurement points. Another parameter that was also taken into account was the ability to predict the experimental spreading rates of both centreline velocity and H_2 mass fraction. The spreading rate is an indicator of the width of the mixing region and is also often used to assess the turbulence models' accuracy (for example [31,32], and [33]). However, as stated in [30], it is only one parameter and it does not provide information about the shape of the profile. Concentration and flow velocity in both experiments and simulations were fitted by Gaussian profiles at the distances from the release that lie within the momentum-dominated regimes and their spreading rates were calculated from the Gaussian fits. The spreading rates were defined as the value of the radial distance from the centreline where H_2 mass fraction and velocity are half their centreline value.

6.1. *Grid sensitivity analysis*

Before evaluating the notional nozzle approaches with the turbulence models, a grid sensitivity analysis was carried out with 4 different mesh resolutions by varying the number of grid points and therefore the grid spacing of a conical geometry starting near the release and extending 2.3 m far from it. The rest of the grid was kept the same for all 4 meshes. The grid resolution at the source was always ¼ of the diameter of the notional nozzle of each approach. The grid characteristics of the 4 meshes, namely coarse, fine, finer and finest are given in the following table. The size of the domain was kept constant (15 m × 10 m × 10 m).

The HD22-24 set was chosen for the sensitivity analysis. All 5 approaches with the 3 turbulence models were simulated using the 4 different grids. A grid independent numerical solution was found with the finer grid, as the difference between the flow velocity and H_2 concentration with the ones of the finest grid was between 2% and 5% for most cases whereas the maximum discrepancy was less than 10%. The same strategy that was used to generate the finer grid for HD22-24 was adopted for the computational meshes for HD35-37 and HD00-01.

Table 4 – Grid characteristics.

	Coarse	Fine	Finer	Finest
Total number of nodes	47.500	68.800	179.300	533.300
Number of nodes in the conical geometry	5.500	19.300	72.100	344.600
Minimum (10^{-4} m) and maximum (10^{-1} m) grid size in the conical geometry	4.8 1.56	4.8 1.1	4.8 0.69	4.8 0.3

6.2. *Simulation results and comparison with experimental data*

Two statistical performance indicators were used to evaluate the approaches, as a direct comparison between the results of the experimental data and simulations was not practical due to the large number of simulations. These statistical indicators were recommended by Chang and Hanna [34] for evaluating air dispersion models. However, they have been used often in different types of scenarios, such as subsonic H_2 jet releases in confined spaces [35], H_2 deflagrations in a tunnel [36], bonfire tests on thermally insulated LPG tanks [37].

Geometric Mean Bias (MG): measures relative mean bias and indicates only systematic errors based on a logarithmic scale. Values of 0.5 and 2.0 can be thought of as 'factor of two' over- and under-predictions in the mean, respectively. It has a value of 1 for an ideal model performance.

$$MG = \exp\left(\overline{\ln V_o} - \overline{\ln V_p}\right) \tag{7}$$

Geometric Mean Variance (VG): measures relative scatter. It has a value of 1 for an ideal model performance.

$$VG = \exp\left[\overline{\left(\ln V_o - \ln V_p\right)^2}\right] \tag{8}$$

where V_o is the observed (experimental) value and V_p is the predicted (numerical) value.

The values of the statistical indicators are aggregated for all 3 measurement points along the centreline (0.75 m, 1.5 m and 2.21 m from the release) for each notional nozzle approach and turbulence model. A graph of MG versus VG is

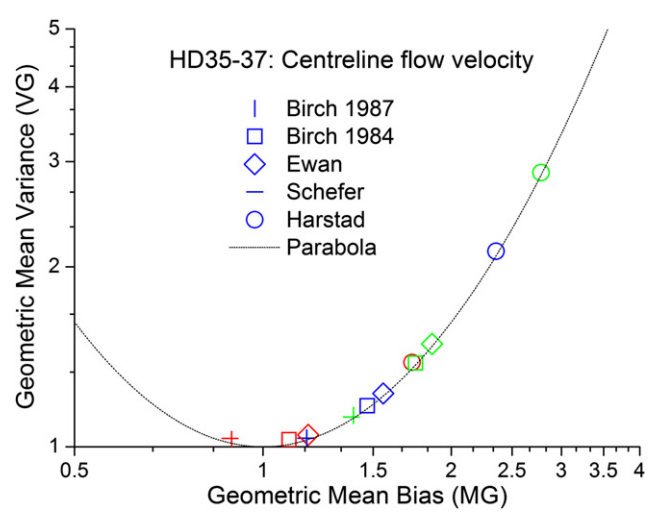

Fig. 2 – Geometric mean bias (MG) versus geometric mean variance (VG) for HD35-37, centreline flow velocity (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

used for a systematic evaluation. This type of graph shows a parabola which represents the minimum possible VG value corresponding to a particular MG value due to systematic bias. All points must lie either on the parabola or inside. Any point located close to the parabola, indicates results with systematic error. The ones close to the central axis ($x = 1$) indicate dispersion of results. Points on the left hand side of the central axis show a tendency to over prediction. The opposite holds for points on the right hand side. For a perfect agreement between numerical results and experimental data, the points should be located on the parabola vertex (1,1).

6.2.1. *Case HD35-37 (H_2 release from 1 mm pipe diameter at 53 bar)*

Fig. 2 shows the values of MG versus VG of the centreline flow velocity of the experimental set HD35-37. In general, most of simulations under predict the experimental value of the centreline velocity except Birch1987 and Schefer with k-ε model. A direct comparison between each simulation and experimental data showed that there is a tendency to under predict the centreline velocity at 1.0 m and 2.25 m except Birch1987 and Schefer with k-ε model which gave results slightly higher than the experimental (the same approaches with the other 2 models produced results slightly lower than the experimental). At the closest point (0.75 m) Birch1984 and Ewan approaches with k-ε model predicted values very close to the experimental whereas the other 2 models showed under prediction. Also, Birch1987 and Schefer with BSL and SST gave results very close to the experimental close to the release. The same approaches with k-ε model overpredicted the data. Harstad approach showed generally the largest under prediction, with BSL and SST being less accurate whereas all three profiles were not symmetrical about the centreline at 2.25 m due to buoyancy effects. Finally, the comparison showed that irrespective of the approach, k-ε model produces the highest centreline values (which is intimately linked to the narrower radial profiles). The other two models had a similar performance with BSL producing always either slightly or moderately higher values at the centreline. As Fig. 2 shows, the under prediction (MG value) is less than 2 for all cases apart from Harstad approach with BSL and SST

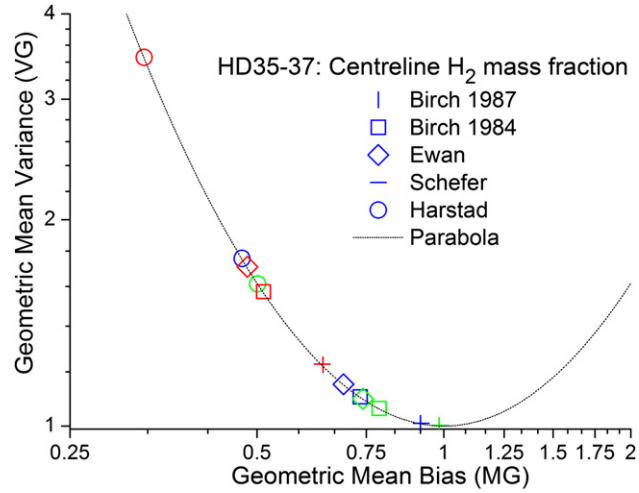

Fig. 4 – Geometric mean bias (MG) versus geometric mean variance (VG) for HD35-37, centreline H_2 mass fraction (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

models which though higher, did not exceed a factor of 3. The fact that all points lie very close, or on the parabola, shows that all simulations have only a mean bias with respect to experimental data, i.e. for each simulation, the ratio C_o/C_p is nearly constant for all 3 measurement points. This is clearly shown in Fig. 3 (left) where the normalized centreline flow velocity is plotted against the equivalent jet initial conditions following the approach of Yüceil and Ötügen [14], as it is explained in paragraph 5. The slope of the straight line fitting the experimental data is only 1.2% lower than the proposed by Chen and Rodi [25]. The slope of Birch1984 approach was 28%, 5% and 54% higher than the one of experimental data for BSL, k-ε and SST models respectively. Birch1987, Ewan and Schefer showed the same tendency, i.e. the closest value of the slope was produced by k-ε (between 5% and 8% overprediction). The other two models produced values by 28%–58% higher. Harstad approach with all turbulence models deviated from the

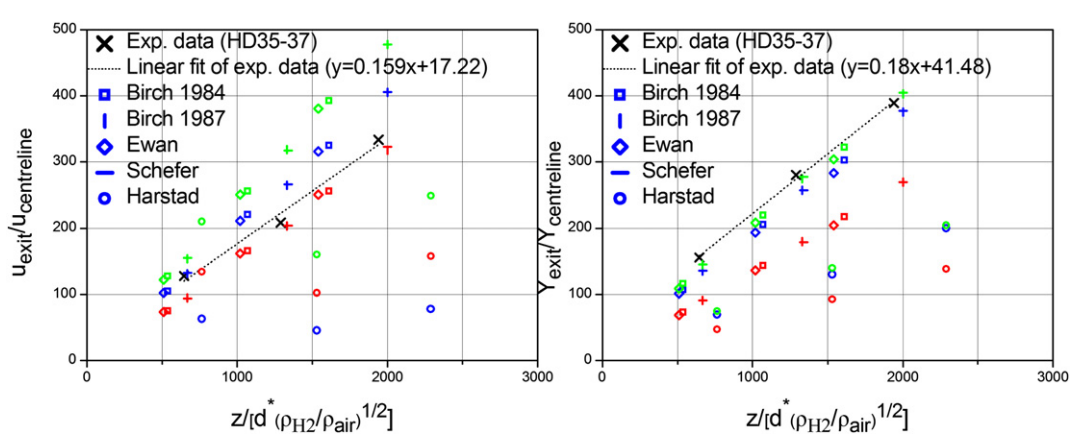

Fig. 3 – Normalized centreline flow velocity (left) and H_2 mass fraction (right) for HD35-37 (green: SST, blue: BSL and red: k-ε), where d* in the x-axis label is the "notional" diameter. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

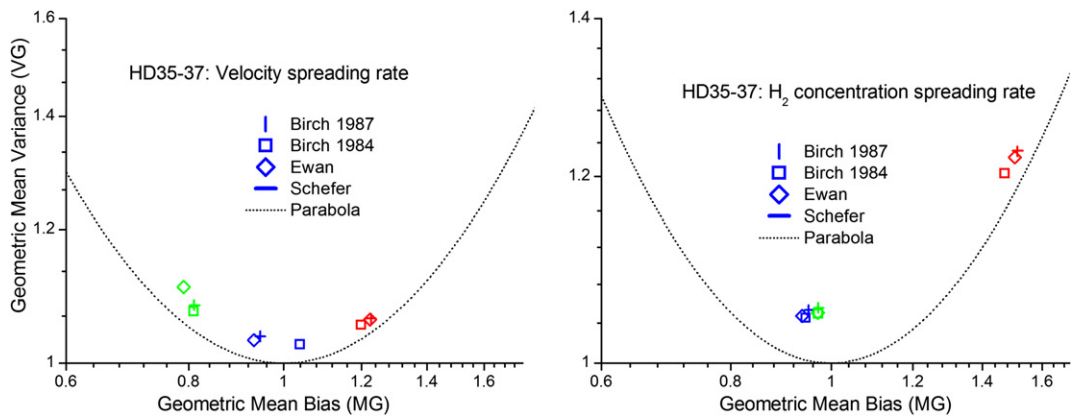

Fig. 5 – Geometric mean bias (MG) versus geometric mean variance (VG) of velocity spreading rate (left) and H_2 mass fraction (right) for HD35-37 (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

experimental value considerably (approximately by −94%), however this was partly expected as the extent of the momentum-dominated regime included only the first measurement point at x = 0.25 m (see Table 3).

The results of the centreline H_2 mass fraction showed in general a tendency to overprediction (Fig. 4). Simulations with Birch1984 and Ewan overpredicted at all 3 points irrespective of the model used with k-ε performing the worst. Birch1987 and Schefer performed very well with BSL, underpredicted slightly with SST and clearly overpredicted with k-ε which deviated from the data the most, at all 3 points. Harstad approach overpredicted considerably at all three points and with all models, with k-ε performing the worst. The profile is again not symmetrical about the centreline at 2.25 m due to buoyancy. As with the flow velocity results, irrespective of the approach used, k-ε model produces the highest centreline values followed by BSL and SST (the last two models performed similarly in most cases). As Fig. 4 shows, the over prediction is in general less than 0.5, apart from Harstad approach which especially with k-ε model was close to 0.3. Again, all points being located on the parabola show that simulations have only a mean bias with respect to experimental data. Fig. 3 (right) shows the normalized centreline H_2 mass fraction. The slope of the straight line fitting the experimental data was 2.5% lower than that proposed in [29]. The slope of Birch1984 approach was 1% and 6% higher than the one fitting the experimental data for BSL and SST and 25% lower for k-ε. Birch1987, Ewan and Schefer showed the same tendency, i.e. the closest value of the slope was produced by BSL and SST models whereas k-ε under predicted the value by approximately 25%. Again, Harstad approach deviated the most irrespective of the model used (−50% to −66%).

Generally, most of the approaches performed well for this experimental set, with Birch1987 and Schefer approaches performing the best when BSL or SST model was used and Harstad approach performing the worst among all (the profiles were again off y-axis at 2.25 m). The k-ε model produced higher centreline values (therefore narrower radial profiles) than the other two models.

Another parameter that was investigated was the ability of the approaches and models to predict the experimental spreading rates of velocity and H_2 mass fraction. Fig. 5 (left) shows that the spreading rate of velocity is predicted more accurately when the BSL model is applied irrespective of the approach, whereas SST clearly over predicts the rate and k-ε clearly under predicts it. However, all points are located close to the parabola's vertex, so all performed rather well. The performance was also good for the H_2 concentration spreading rate (Fig. 5, right). This time both BSL and SST performed very well, whereas k-ε under predicted by a factor not higher than 1.5 (MG). Simulations with Harstad approach were not included as, based on the scaled distance x_b, only the nearest to the release point was located within the momentum-dominated regime (see Table 3 where x < 1.2 m) and therefore a Gaussian fit wouldn't be appropriate for fitting the profiles at the other 2 measurement points.

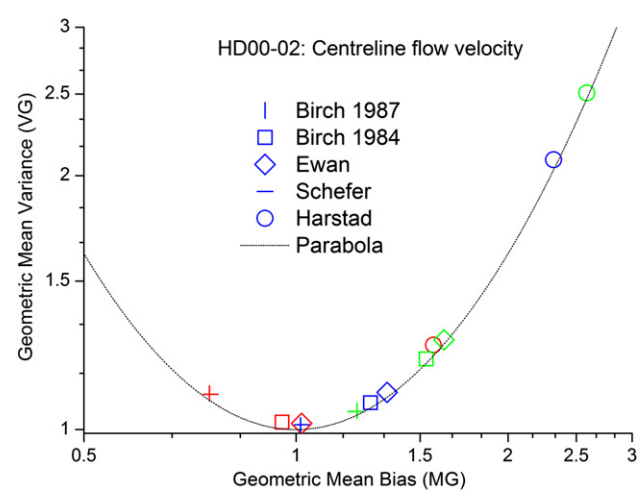

Fig. 6 – Geometric mean bias (MG) versus geometric mean variance (VG) for HD00-02, centreline flow velocity (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

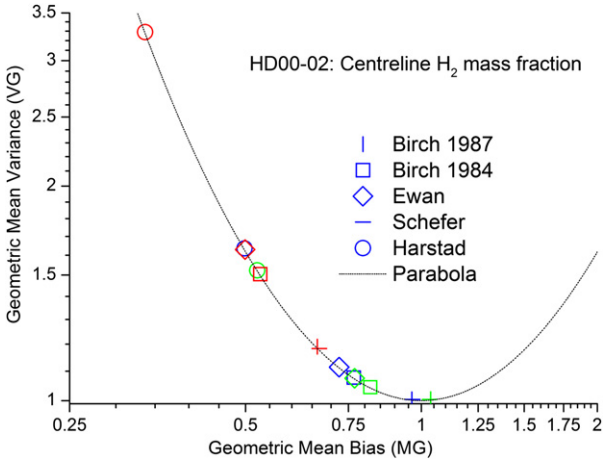

Fig. 7 – Geometric mean bias (MG) versus geometric mean variance (VG) for HD00-02, centreline H_2 mass fraction (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6.2.2. *Case HD00-02 (H_2 release from 0.75 mm pipe diameter at approximately 110 bar)*

Fig. 6 shows the values of MG versus VG of the centreline flow velocity for the HD00-02 set. As in previous experimental set, there is generally a tendency to underprediction. Specifically, Birch1984 and Ewan underpredicted with BSL and SST at all measurement points. The same approaches with k-ε overpredicted close to the release and underpredicted moderately at the other two measurement points. Birch1987 and Schefer approaches performed well at all points especially when BSL or SST model was used (the approaches overpredicted the data with k-ε). Harstad approach underpredicted the data irrespective of the turbulence model applied with BSL and SST producing the lowest values. As in the previous set, most of the simulations produced MG values well within the range $0.5 < MG < 2$. Fig. 7 shows the values of MG versus VG of the centreline H_2 mass fraction. As in previous set, there is a general tendency to over-prediction. Again, Harstad approach performed the worst, overpredicting irrespective of the model applied. The same holds for Birch1984 and Ewan approaches, with k-ε performing the worst whereas the results with BSL and SST were close to the experimental. Birch1987 and Schefer performed again rather well, especially when BSL and SST models were applied. The over-prediction (MG) was not below 0.5 with the exception of Harstad approach (the highest over-prediction was produced with k-ε, see Fig. 7).

Generally, the results of flow velocity and H_2 mass fraction had the same trend as in the previous experimental set: Birch1987 and Schefer approaches performed the best when BSL or SST model was used and Harstad performed the worst. Concerning the turbulence models used, all simulations with k-ε produced the highest values, followed by BSL and SST.

Fig. 8(left) shows the normalized centreline flow velocity. The slope of the straight line fitting the experimental data is 42% higher than the proposed in [26]. The slope of Birch1984 approach deviated from the experimental by −11%, −27% and 6% for BSL, k-ε and SST respectively. Birch1987, Ewan and Schefer approaches showed the same tendency, i.e. approximately −10% to −14%, −26% to −28% and 4%–7% for BSL, k-ε and SST respectively. Harstad approach produced lower slopes by 41%–66%. Fig. 8 (right) shows the normalized centreline H_2 mass fraction. The slope from the experimental data is 10% lower than the theoretical [29]. The slope of the approaches deviated by −3% to −5%, −28% to −30% and −0.3% to 5% for BSL, k-ε and SST respectively. Harstad approach produced lower, than the experimental, slopes by 52%–71%.

Fig. 9 shows the MG versus VG plots for the spreading rate of velocity (left) and H_2 mass fraction (right). Concerning the velocity spreading rate, it seems that k-ε model performs better than the other two which over predict the rate. However, the over prediction is not less than 0.6. The performance was also good for the H_2 concentration spreading rate with the k-ε model under predicting the rate by a factor not higher than 1.5 and BSL and SST models over predicting by a factor not less than 0.85.

6.2.3. *Case set HD22-24 (H_2 release from 0.25 mm pipe diameter at approximately 160 bar)*

Fig. 10 and Fig. 11 show the values of MG versus VG of the centreline flow velocity and H_2 mass fraction of HD22-24 set respectively. This time all simulations overpredicted the

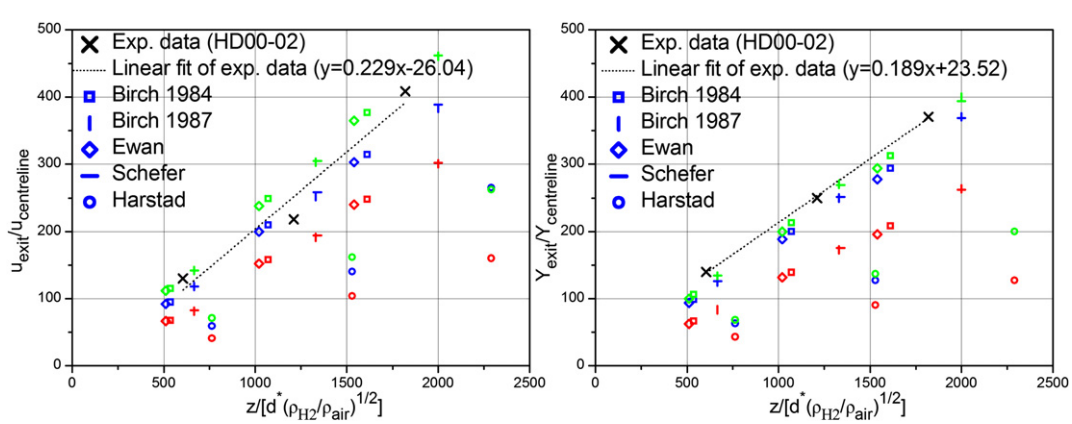

Fig. 8 – Normalized centreline flow velocity (left) and H_2 mass fraction (right) for HD00-02 (green: SST, blue: BSL and red: k-ε); where d^* is the "fictitious" diameter. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

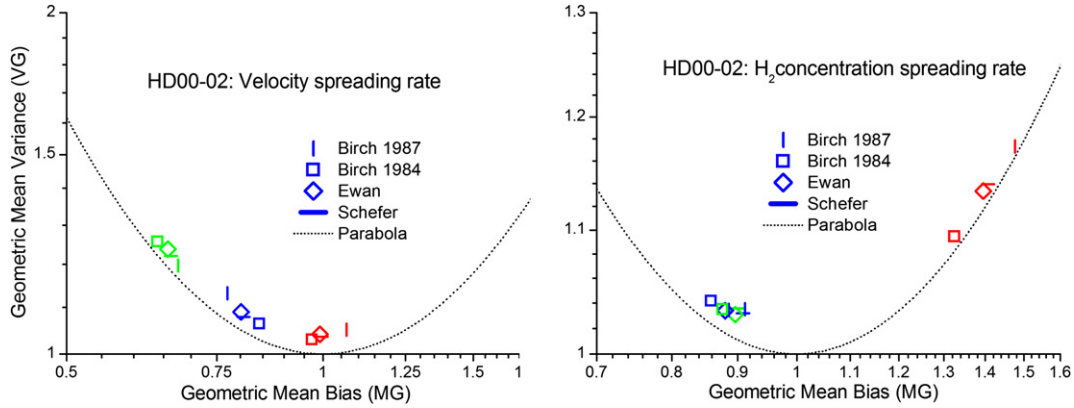

Fig. 9 – Geometric mean bias (MG) versus geometric mean variance (VG) of velocity spreading rate (left) and H2 mass fraction (right) for HD00-02 (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

velocity at the closest point. Concerning the other two points, Birch1987 and Schefer overpredicted irrespective of the model used with k-ε performing the worst whereas Birch1984 and Ewan overpredicted with k-ε, slightly underpredicted with BSL and ST. Concerning the H_2 mass fraction predictions, all simulations overpredicted with Birch1984 and Ewan performing the worst (especially with k-ε).

Fig. 12 shows the normalized centreline velocity and H_2 mass fraction. The slope of the lines fitting the experimental data is 160% and 34% higher than the theoretical ([26] and [29]) for velocity and H_2 mass fraction respectively. The slopes from simulations were lower than the experimental by 34%–57% and 20%–48% for velocity and H_2 mass fraction. Harstad approach produced profiles that were off the release axis at both 1.5 m and 2.25 m.

6.3. *Discussion*

The results are mainly discussed and assessed in terms of the physical (real) nozzle diameter, the notional nozzle approach and the turbulence model applied. Also, both simulations and experimental data are compared with the centreline decay laws proposed in literature.

A comparison of simulations with the three different nozzle diameters applying the same notional nozzle approach and turbulence model showed that the performance is getting moderately worse with decreasing nozzle diameter. Also, there was a clear tendency of H_2 mass fraction overprediction in all experimental sets (with results of HD22-24 set -smallest physical diameter- deviating 1.3–3 times from the data). On the other hand, the flow velocity of HD35-37 (1 mm diameter) and HD00-02 (0.75 mm diameter) sets was underpredicted but

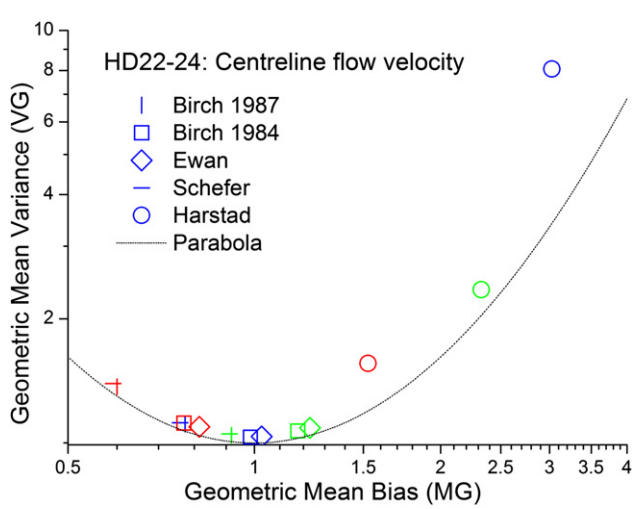

Fig. 10 – Geometric mean bias (MG) versus geometric mean variance (VG) for HD22-24, centreline flow velocity (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

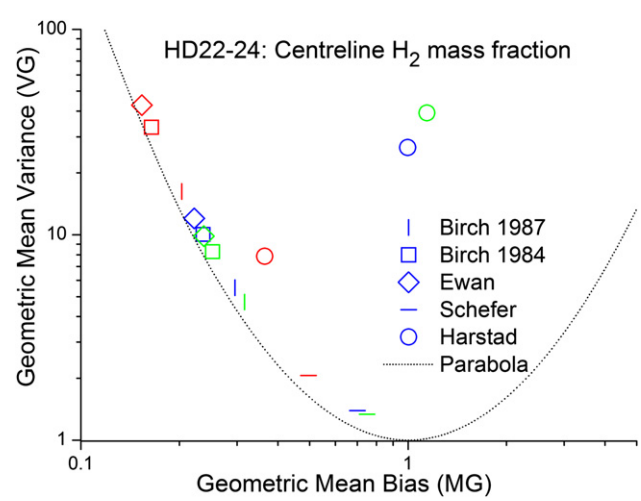

Fig. 11 – Geometric mean bias (MG) versus geometric mean variance (VG) for HD22-24, centreline H_2 mass fraction (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

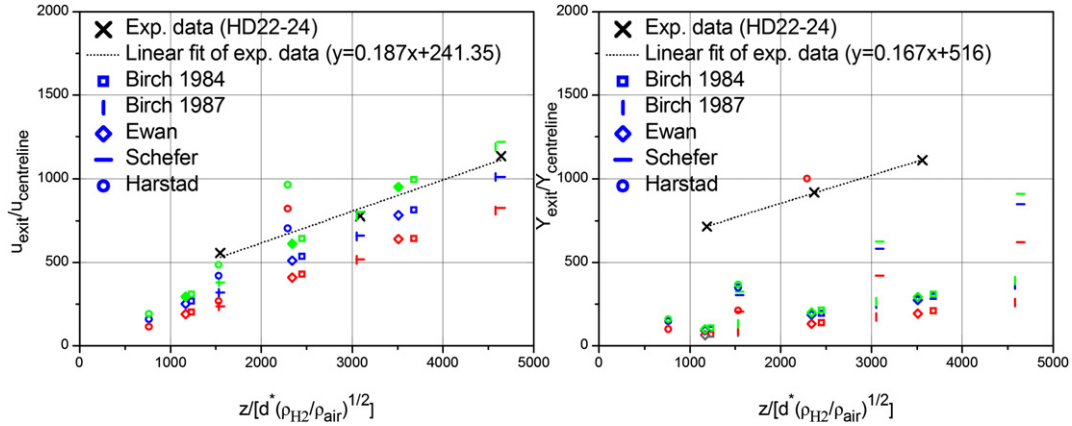

Fig. 12 – Normalized centreline flow velocity (left) and H_2 mass fraction (right) for HD22-24 (green: SST, blue: BSL and red: k-ε). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

this tendency was not clearly followed in HD22-24 set (0.25 mm diameter).

Concerning the notional nozzle approaches, the results can be classified into three groups having broadly a similar performance, one group with Birch1984 and Ewan, another with Birch1987 and Schefer and finally Harstad alone. Generally, Birch1987 and Schefer approaches performed better, followed by Birch1984 and Ewan and lastly Harstad which deviated from the data the most. This grouping can be associated with the conditions at the notional nozzle of each approach. From Table 2 (which gives the conditions at each notional nozzle) it can be seen that the momentum flow rate and net rate of energy flow of Birch1987 and Schefer are always higher and similar, followed by Birch1984 and then closely by Ewan and finally Harstad.

Concerning the overall performance of turbulence models for H_2 concentration predictions, it seems that SST model is better followed closely by BSL and lastly k-ε. For the flow velocity, the conclusions are not as straightforward. It seems that k-ε performs better with Birch1984 and Ewan (especially for the larger release diameter) whereas SST and BSL are closer to the experimental data with Birch1987 and Schefer approaches.

One could expect SST to perform similar to k-ε as it is widely known that SST switches to a k-ε model (actually this k-ε model is transformed back to a set of k-ω equations by using a blending function) in free shear jet calculations. However, there are differences between the k-ε models that can justify the different performance found in this study. As mentioned in [38] the differences between this model and the exact transformation of the ε-equation of the standard k-ε to an ω-equation results in an extra diffusion term that is not included in the SST model. Also, as reported in [39], the ω-equation diffusion coefficient has a value of $\sigma_{\omega 2} = 1/\sigma_\varepsilon = 1/1.3 = 0.769$, whereas in SST $\sigma_{\omega 2}$0.857 which corresponds to $\sigma_\varepsilon = 1.17$. Focussing in k-$\varepsilon$, its performance in round jet calculations has been commented in the literature. For example in [23] it has been stated that "a well known deficiency of the k-ε model is that it significantly overpredicts the rate of spreading for the round jet". Also, in [31] and [40] the authors presented the performance of four eddy viscosity turbulence models (among them the k-ε of Launder and Sharma, k-ω of Wilcox and SST of Menter). All models were found to overpredict the thickness of the experimental velocity profile (i.e. wider profiles) and this was attributed to the "classical well known anomaly in these models that have been fine-tuned with empirical data of mixing layer, plane jet and/or far wake experiments". Having in mind the intimate linkage between half-widths (thickness of profile) and centreline decay in the absence of buoyancy [41], the results of this study show clearly the same tendency for the first two experimental sets (systematic under-prediction of centreline velocity).

Finally, the decay laws of centreline velocity and H_2 mass fraction of both experimental data and simulations were not universal. Concerning the experiments, the slopes were very close to the theoretical value for the experimental set with the larger diameter pipe and the lower stagnation pressure. The slopes assumed different values for the other two experiments. Even though there are several experimental studies that suggest universal asymptotic values for the region sufficiently far downstream the nozzle (such as in [26,42], and [43]), there are works that suggest that turbulent flows can asymptote to a variety of self-similar states determined by their initial conditions (for example in [44,45], and [46]). Concerning the simulations, the slopes deviated from the experimental values no more than 58% and 25% for velocity and H_2 mass fraction for the first two sets (excluding Harstad approach) whereas for the one with the smallest diameter pipe the slopes deviated substantially from the experimental ones.

7. Conclusions

An evaluation of selected notional nozzle approaches (Birch1984, Birch1987, Ewan, Schefer and Harstad) for CFD simulations of free-shear under-expanded round jets was presented. Three turbulence models were applied for each notional nozzle approach (BSL, k-ε and SST) and the results were compared with 3 experimental sets of high pressure H_2 releases from small pipes.

An attempt to justify and explain the differences found in the notional nozzle approaches and turbulence models was presented. It was found that for the same turbulence model and notional nozzle approach, the performance was gradually moderately deteriorating with decreasing nozzle diameter. Comparing the notional nozzle approaches alone, it was found that these could be grouped based on the momentum and energy flux at the source with Birch1987 and Schefer generally performing better, followed by Birch1984 and Ewan and lastly Harstad. Evaluating the turbulence models, it was found that the SST performed better followed closely by BSL and lastly k-ε for prediction of H_2 concentration. For the flow velocity, the k-ε model performs better with Birch1984 and Ewan (especially for the larger release diameter) while SST and BSL are closer to the experimental data with Birch1987 and Schefer approaches.

Experiments of highly under-expanded buoyant jets with measurements closer to the release and from different release diameters and pressures should be studied numerically in the future using notional nozzle approaches together with different turbulence models in order to make more general conclusions about their performance. The question about the universality or the range of applicability of centreline decay laws and the value of their constants concerning highly under-expanded buoyant jets should also be addressed.

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